

## Analysis First-Year Exam, May 2025, Part 1

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Answer *FOUR* questions in detail. State carefully any results used without proof.

1. Let  $A$  be a subset of the metric space  $M$ . Decide (with proof or counterexample) whether the following statements are equivalent:
  - (i)  $A$  is open;
  - (ii) for each  $X \subseteq M$ , if  $A \cap X = \emptyset$  then  $A \cap \overline{X} = \emptyset$ .
2. The function  $f : X \rightarrow Y$  between metric spaces is distance-preserving. Must  $f(X)$  be closed in  $Y$  in each of the following cases? If the answer is Yes, give a proof; if the answer is No, give a counterexample.
  - (i)  $X$  is compact.
  - (ii)  $X$  is complete.
3. Let  $f : X \rightarrow Y$  be a continuous bijection,  $g : Y \rightarrow X$  its inverse. Must  $g$  be continuous in each of the following cases? Proof or counterexample, as appropriate.
  - (i)  $X$  is compact.
  - (ii)  $Y$  is compact.
4. The function  $f : \mathbb{R} \rightarrow \mathbb{R}$  is continuous and satisfies

$$\liminf f(n) < 0 < \limsup f(n)$$

where  $n$  runs through the natural numbers. Prove that each of  $f^{-1}(-\infty, 0)$  and  $f^{-1}(0, \infty)$  is infinite.

5. Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be such that

$$|f(x) - f(y)| \leq |x - y|^a$$

for some  $a > 1$  and all  $x, y \in \mathbb{R}$ . Deduce as much as possible about  $f$ .